NIKITA SIZYKH



Experimental Physicist · Semiconductor & Ferroelectric Devices

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PROFILE

Experimental physicist and microelectronics R&D researcher with end-to-end experience in thin-film device fabrication, electrical characterization, synchrotron-based spectroscopy (HAXPES, DAFS), data analysis, and physics-based modeling. Research focuses on HfO_2–ZrO_2 ferroelectric devices, reliability, electrically active defects, and buried interfaces examination. Experienced in developing custom experimental setups, automated measurement workflows, and operando synchrotron experiments.

EXPERIENCE

Moscow Institute of Physics and Technology (MIPT)

Research Engineer → Junior Research Fellow | 2020–2024 → 2024–present
Lab of Functional Materials and Devices for Nanoelectronics

Ferroelectric HZO devices for non-volatile memory

2022–2026

Developed and fabricated HZO-based devices and test structures on standalone chips and 4″ wafers, including refinement of fabrication processes. Investigated polarization switching, retention loss, endurance, leakage, dielectric breakdown and multilevel operation; developed automated electrical measurement and data-processing workflows.

Operando DAFS/NEXAFS of electrically biased devices

2024–2025

Developed and implemented the electrical part of operando synchrotron experiments and post-processing; participated in integration of the experimental setup at the synchrotron end-station.

HAXPES metrology and standardization for synchrotron measurements

2021–2022

Contributed to the development of a certified measurement procedure for determining the electronic and chemical properties of functional thin-film structures using HAXPES, within a national program on measurement traceability at synchrotron-radiation facilities.

Buried metal/HZO interface characterization by NTR SW-HAXPES

2021–2022

Developed an analysis methodology combining NTR SW-HAXPES with X-ray reflectivity modeling to investigate the chemical and electronic structure of buried metal/HZO interfaces, refine the multilayer structure, and determine Schottky barrier heights.

JSC NIIGraphite

Research Engineer | 2019–2021
Department of Advanced Development

Raman spectroscopy of SiC-based composite materials

2019–2020

Participated in the development of a Raman-spectroscopy-based methodology for assessing component distribution in silicon-carbide-based composite materials.

EDUCATION

PhD Candidate, Condensed Matter Physics

Moscow Institute of Physics and Technology | 2022–2026
Thesis defense scheduled for December 2026

Thesis: “Study of Interfaces in Ferroelectric Hafnium–Zirconium Oxide-Based Capacitor Structures for Memory and Neuromorphic Applications”

MSc, Applied Mathematics and Physics

Moscow Institute of Physics and Technology | 2020–2022

Thesis: “Investigation of the Chemical and Electronic Properties of the Buried HZO/W Interface Utilizing NTR SW-HAXPES”

BSc, Nanoengineering

Bauman Moscow State Technical University | 2016–2020

NIKITA SIZYKH

Technical expertise · publications · additional experience

SKILLS

Device fabrication & processing

MIM, MFM, MFMIS, and MOS structures; magnetron sputtering; e-beam evaporation; optical lithography with positive and negative photoresists; wet and dry etching.

Electrical characterization & reliability

Probe stations equipped with a semiconductor device parameter analyzer; custom test benches using SMUs, WGFMUs, AWGs, and OSCs; I–V and C–V characterization; ferroelectric switching, retention, endurance, electrical stress and dielectric breakdown; experiment automation.

Materials characterization

XPS; NTR SW-HAXPES; XRD; X-ray reflectivity; Operando DAFS; SEM; FIB cross-sectional lamella preparation for TEM and Slice-and-View.

Scientific Computing, Automation & Engineering Tools

Python; Matlab; C/C++ (Arduino); instrument control (VISA); scientific data processing; physics-based modeling; Git; KLayout; Altium Designer; Autodesk Eagle; Autodesk Fusion 360.

Languages & Scientific Communication

Russian — Native; English — B2; manuscript preparation and internal review; preparation of responses to reviewers; conference presentations.

PUBLICATIONS

Interface defects as the origin of breakdown in ferroelectric TiN/HZO/TiN memory capacitors subjected to electrical stress

N. A. Sizykh, M. Spiridonov, A. Khanas, A. V. Zenkevich

Applied Physics Letters 128, 252903 (2026) · DOI 10.1063/5.0336988

How to make opposite state store again

D. S. Kuzmichev, V. S. Konstantinov, **N. A. Sizykh**, R. R. Khakimov

Nanosystems: Physics, Chemistry, Mathematics 17(2), 187–192 (2026) · DOI 10.17586/2220-8054-2026-17-2-187-192

Realization of Deterministic Multilevel Polarization States in Ferroelectric $Hf_{0.5}Zr_{0.5}O_2$-Based Capacitors

N. A. Sizykh, M. Spiridonov, A. Khanas, A. V. Zenkevich

Russian Microelectronics 54(8), 886–891 (2025) · DOI 10.1134/S1063739725601808

Short-Term Bienenstock-Cooper-Munro Learning in Optoelectrically-Driven Flexible Halide Perovskite Single Crystal Memristors

I. Matchenya, A. Khanas, R. Podgornyi, D. Shirkin, A. Ekgardt, **N. A. Sizykh**, et al.

Small Methods 9(9), e00203 (2025) · DOI 10.1002/smtd.202500203

TiN/HZO/TiN ferroelectric capacitors with TiO_2 insets: Critical difference between top and bottom interface modification

A. G. Chernikova, **N. A. Sizykh**, I. V. Zabrosaev, A. M. Markeev

Surfaces and Interfaces 62, 106135 (2025) · DOI 10.1016/j.surfin.2025.106135

Effect of Domain Structure and Dielectric Interlayer on Switching Speed of Ferroelectric $Hf_{0.5}Zr_{0.5}O_2$ Film

A. Chouprik, E. Savelyeva, E. Korostylev, E. Kondratyuk, S. Zarubin, **N. A. Sizykh**, et al.

Nanomaterials 13(23), 3063 (2023) · DOI 10.3390/nano13233063

Effect of Electrode Nanopatterning on the Functional Properties of Ta/TaO_x/Pt Resistive Memory Devices

M. Yu. Zhuk, **N. A. Sizykh**, D. S. Kuzmichev, A. A. Chouprik, S. S. Zarubin, E. S. Gornev, A. V. Zenkevich

ACS Applied Nano Materials 5(6), 8594–8601 (2022) · DOI 10.1021/acsanm.2c02076

AWARDS

Best Oral Presentation

Synchrotron Radiation Techniques for Catalysts and Functional Materials · Tomsk · 2024

“Application of hard X-ray photoelectron spectroscopy in near total reflection regime for the study of nanoelectronic devices”

ROSATOM Award · “Innovative Scientific Leader in Nuclear Field”

2020

ADDITIONAL EXPERIENCE

Web Dashboard for Users of MIPT Center of Shared Research Facilities

2024-2026

Designed the dashboard architecture and maintained the service on a daily basis.

FIB–SEM characterization of reservoir rock cores

2022-2023

Prepared cross-sections of reservoir rock core samples using focused ion beam (FIB) milling; generated SEM image stacks for 3D reconstruction of pore-space morphology and fluid flow simulations.

Physics tutor

2019-2022

Tutored high-school and university students in physics and prepared them for examinations.